From Gaussians to Bayesian optimisation

From the bell curve to choosing the next experiment

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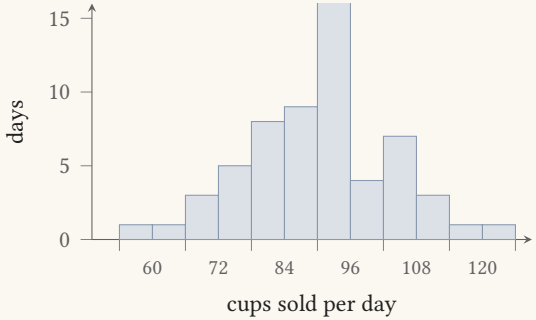
21 October 2026

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Stage 1

The 1D Gaussian

Daily sales, and a question



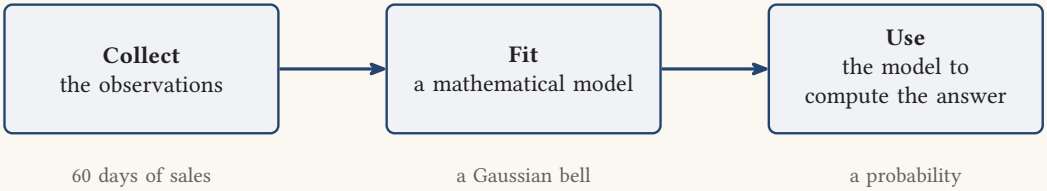
Real-world observations from 60 days.

What is the chance that tomorrow we sell **between 90 and 120 cups**?

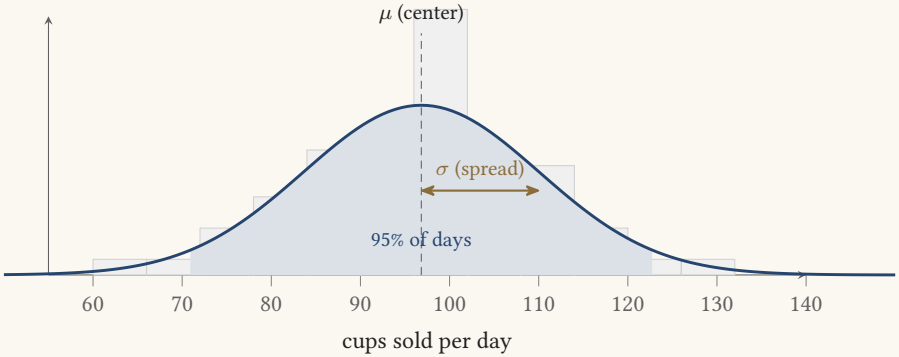
Prepare too little and we run out. Prepare too much and it melts.

$$P(90 \leq X \leq 120) = ?$$

How data science answers it

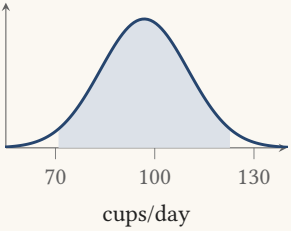


The model is a Gaussian



$$g(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

Estimating the center and the spread



the fitted bell

Average of the data, which settles at the true mean:

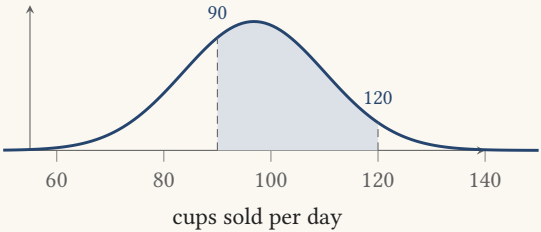
$$\hat{\mu} = \frac{1}{n}\sum_{i=1}^{n} x_i \xrightarrow[n\to\infty]{} \mathbb{E}[X] =: \mu$$

Average squared distance from the center, i.e. the variance:

$$\hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n} (x_i - \hat{\mu})^2 \xrightarrow[n\to\infty]{} \mathbb{E}\big[(X-\mu)^2\big] =: \sigma^2$$

$$\hat{\mu} = 96.8 \qquad \hat{\sigma}^2 = 174.8$$

Answering the question



$$P(90 \le X \le 120) = \int_{90}^{120} g(x)\,dx$$

$$P = 0.66$$

about a 2-in-3 chance

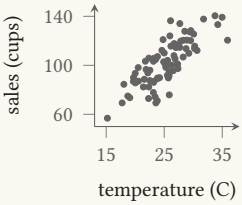
observations → a Gaussian model → a probability we can plan against

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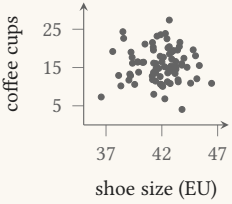
Stage 2

The 2D Gaussian

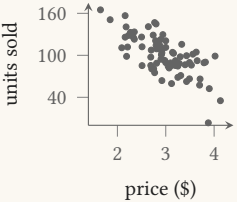
Three worlds, and a question



hotter → more sales



no relationship



pricier → fewer sold

On the first world, what is the chance a day is **warm and busy at once**?

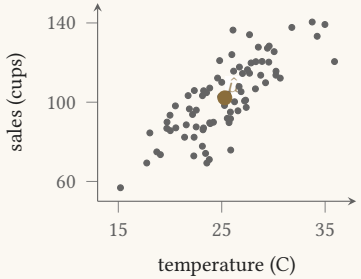
temperature between 25 and 30 C, sales between 100 and 140 cups

$$P\big(\, T \in [25, 30] \;\text{ and }\; S \in [100, 140] \,\big) \;=\; ?$$

The same method, now in 2D



Estimating the pieces: center and per-axis spread



the mean, marked

Idea: run Stage 1 once on each axis.

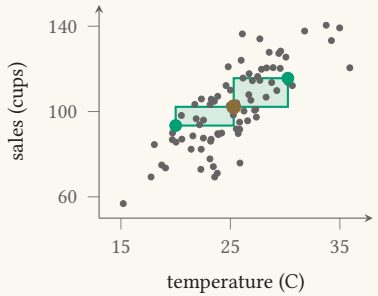
The mean vector (average of each coordinate):

$$\hat{\boldsymbol{\mu}} = \begin{pmatrix} \hat{\mu}_1 \\ \hat{\mu}_2 \end{pmatrix} = \begin{pmatrix} 25.3 \\ 102.2 \end{pmatrix}$$

The spread on each axis (variance):

$\hat{\sigma}_1^2 = 16.0$ (temperature) $\quad\quad$ $\hat{\sigma}_2^2 = 348.2$ (sales)

Estimating the pieces: the correlation



two sample rectangles; signed area = a product

Average of those signed areas:

$$\hat{\sigma}_{12} = \frac{1}{n}\sum_{i=1}^{n}(x_{1i} - \hat{\mu}_1)(x_{2i} - \hat{\mu}_2)$$

$$\xrightarrow[n\to\infty]{} \mathbb{E}\big[(X_1 - \mu_1)(X_2 - \mu_2)\big]$$

Its sign is the headline, across the three worlds:

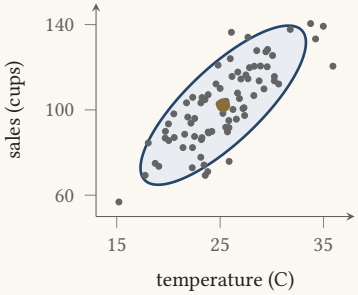
positive $\hat{\sigma}_{12} = +58.8$, corr $+0.79$

neutral $\hat{\sigma}_{12} \approx 0$, corr 0.00

negative $\hat{\sigma}_{12} = -9.7$, corr -0.70

The scale-free reading is the correlation, always in $[-1, 1]$.

The bivariate Gaussian, and its covariance matrix



the fitted bell, tilted by + correlation

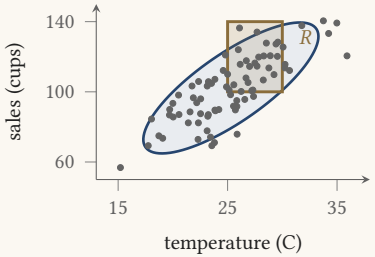
with the point $\mathbf{x} = (x_1, x_2)^\top$ and center $\boldsymbol{\mu} = (\mu_1, \mu_2)^\top$:

$$g(\mathbf{x}) = \frac{1}{2\pi\sqrt{|\Sigma|}} \exp\Big(-\tfrac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^\top \Sigma^{-1}(\mathbf{x}-\boldsymbol{\mu})\Big)$$

Σ is the covariance matrix: spreads on the diagonal, covariance off it.

$$\Sigma = \begin{pmatrix} \sigma_1^2 & \sigma_{12} \\ \sigma_{12} & \sigma_2^2 \end{pmatrix} = \begin{pmatrix} 16.0 & 58.8 \\ 58.8 & 348.2 \end{pmatrix}$$

The answer: a double integral over the region



the region $R = [25, 30] \times [100, 140]$

Chance the day lands inside the rectangle R:

$$P\big((X_1, X_2) \in R\big) = \iint_R g(x_1, x_2)\, dx_1\, dx_2$$

$$P = 0.31$$

about a 1-in-3 chance

two questions at once → a bivariate Gaussian (μ, Σ from data) → a probability over a region

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Stage 3

The multivariate Gaussian

Three flavors over 40 days, and a question

day	vanilla	chocolate	strawberry
1	55	41	35
2	64	44	36
3	41	29	30
4	67	36	39
5	59	41	35
6	73	53	37
⋮	⋮	⋮	⋮

Each number is cups sold that day. 40 days in total.

Each row is one day; each number is cups sold. Hot days lift all three flavors together.

We plan a target range for each flavor. Call a day **balanced** when all three land inside their target at once, so nothing runs out and nothing is left over:

vanilla 40 to 60, chocolate 30 to 50, strawberry 20 to 40 cups

$$P(\text{a balanced day}) = ?$$

The same method, in three dimensions



A mean vector and a covariance matrix

The mean vector, one average per flavor (per column of the table):

$$\hat{\boldsymbol{\mu}} = \big(\,53.4,\;\; 41.5,\;\; 31.4\,\big)$$ (vanilla, chocolate, strawberry), in cups

The covariance matrix $\hat{\Sigma}$ (spreads on the diagonal):

	vanilla	chocolate	strawberry
vanilla	91.6	66.7	51.8
chocolate	66.7	108.0	49.6
strawberry	51.8	49.6	51.1

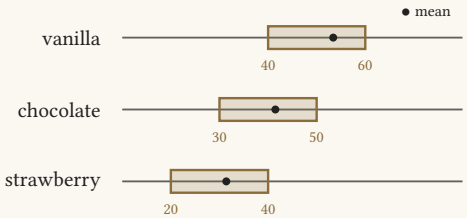
all nine entries positive: the flavors move together.

The general Gaussian density, for a point $\mathbf{x}$ in d dimensions:

$$g(\mathbf{x}) = \frac{1}{(2\pi)^{d/2}\sqrt{|\Sigma|}}\;\exp\!\Big(-\tfrac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^{\top}\Sigma^{-1}(\mathbf{x}-\boldsymbol{\mu})\Big)$$

Here $d = 3$. (Stage 1 was $d = 1$, Stage 2 was $d = 2$; only the size of $\boldsymbol{\mu}$ and Σ changes.)

The answer: a multi-dimensional integral



each bar is one flavor; the shaded band is its target range.

Chance every flavor lands in its target range at once (a balanced day):

$$P(\mathbf{X} \in R) = \iiint_R g(x_1, x_2, x_3)\, dx_1\, dx_2\, dx_3$$

the balanced region $R = [40, 60] \times [30, 50] \times [20, 40]$ cups

$$P = 0.48$$

about half our days are balanced; the rest miss on a flavor

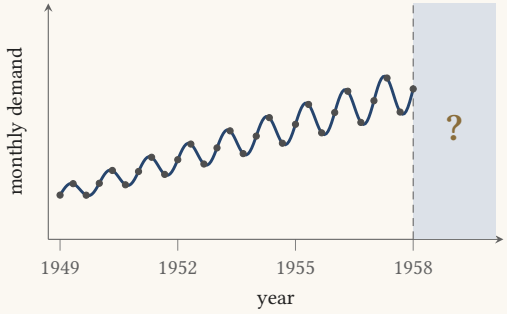
many quantities at once → a multivariate Gaussian ($\boldsymbol{\mu}$ vector, Σ matrix) → a probability over a region

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Stage 4

The Gaussian process

Demand through 1958, and a question



Monthly demand through 1958; the next two years are blank.

We track a monthly **demand** series through 1958.

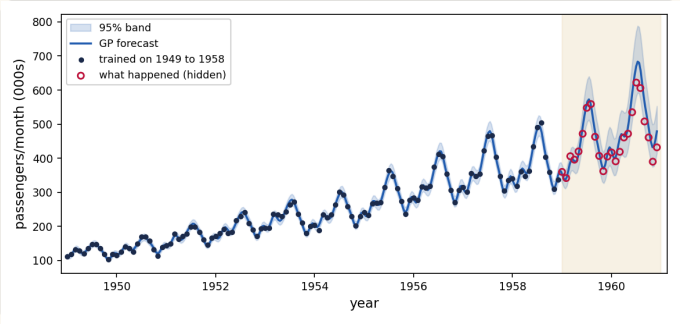
Real airline-passenger numbers stand in for a shop's revenue series.

What will demand be over the **next two years**?

And how **sure** can we be?

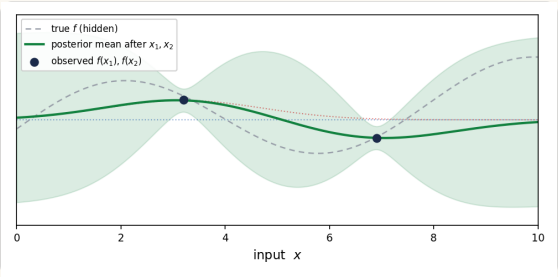
We want an honest range, not just a single number.

Fit a Gaussian process: the forecast



A forecast that also says how sure it is.

So what is a Gaussian process?



Pick x_2, observe $f(x_2)$: the mean **threads both**, and the band tightens. More data, less doubt.

A Gaussian process is just **two** things:

1. A mean function $m(x) = \mathbb{E}[f(x)]$

the best guess at every input (flat before data).

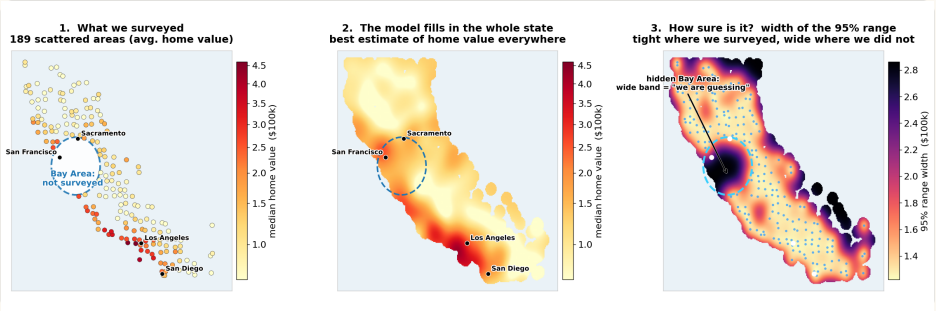
2. A kernel $k(x, x')$

how similar two inputs are; nearby inputs → correlated, so the mean bends and the band pinches at the data.

The **band** is the leftover uncertainty: small at the data, wide away.

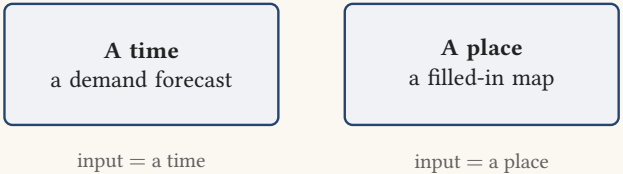
We infer m and k by Bayesian inference, which we do not cover here; the library does it.

The same idea, over a map



The same Gaussian process, a different input space.

One idea, many input spaces



In code it is a few lines: choose a kernel, fit, then predict with a band.

a mean function + a kernel → predict anywhere, with honest error bars

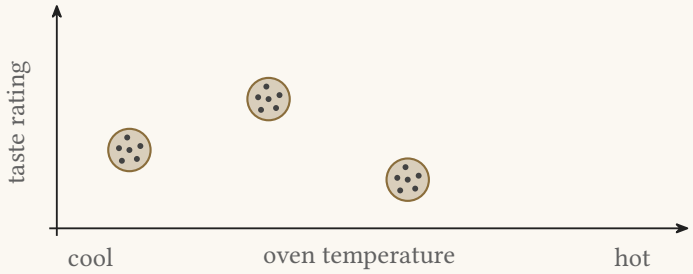
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Stage 5

Bayesian optimisation

Slow bakes, few tries

Finding the best oven temperature for cookies.



1. Each cookie: one bake, one rating.

2. Between them: unknown.

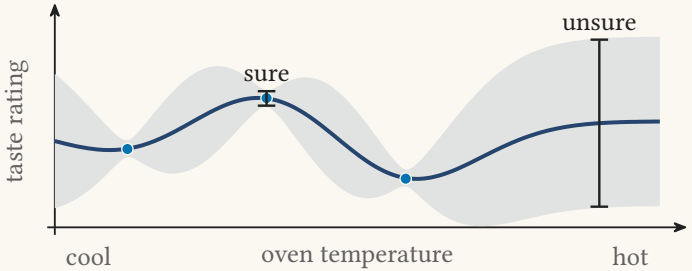
3. A sweep fixes every temperature in advance.

Let each taste guide the next bake.

Kochanski et al. 2017; toy: ours

A cheap model learns from bakes

Ask it about untried temperatures.



1. Line: the prediction.

2. Band: how unsure.

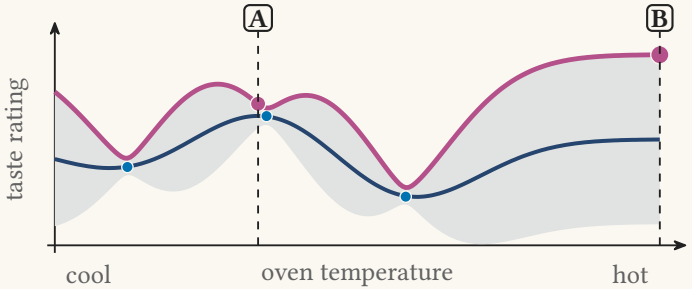
Surrogate model: a cheap stand-in, often a Gaussian process.

It predicts and says how sure.

Frazier 2018; Krause and Hübotter; toy: ours

Try the promising, or the unknown

Two reasons to try a temperature.



Acquisition function: a score per setting.

1. A exploits: the best guess.

2. B explores: the widest band.

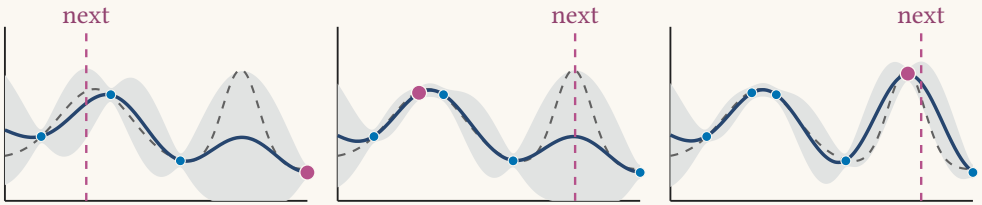
3. Score = top of the band.

Bake at the highest score.

Krause and Hübotter; Brochu et al. 2010; toy: ours

Fit, pick, bake, repeat

One search, three rounds. Dashed: the true curve, unseen.



1. Hottest oven: burnt. **2.** Beside the best: same. **3.** In between: best yet.

This loop is Bayesian optimisation.

Brochu et al. 2010; Shahriari et al. 2016; toy: ours